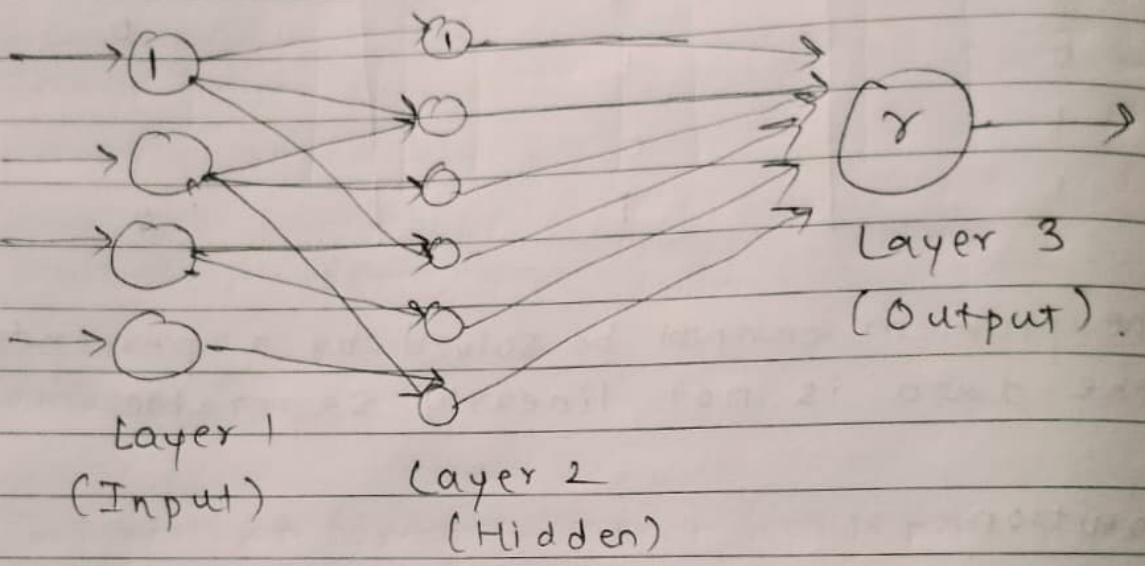
Mod 2

Multi-Layered Neural Networks.

A multi-layered neural network is a Multi-layered perceptron.



~~Layers marke~~

Nodes marked as '1' are bias units.

$$r = \sum_{i=1}^{m} w_i x_i + \text{bias}.$$

for hidden layer,

$$h_1 = F(r)$$

Activation Function

It is a mathematical function applied to the output of a neuron. It introduces non linearity into the model, allowing to learn complex patterns.

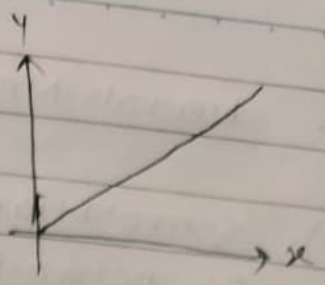
Types

Date :

1) Linear Activation Function

$y = x$

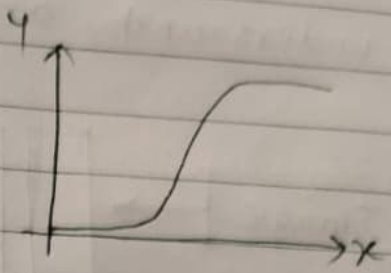
output from $-\infty$ to ∞



2) Sigmoid function

$$y = \frac{1}{1+e^{-x}}$$

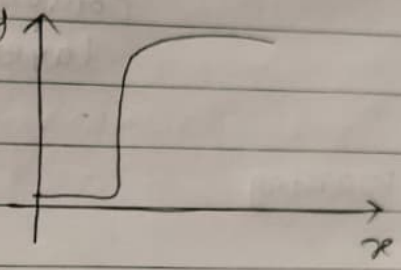
output from -2 to 2



3) Tanh function

$$y = \frac{2}{1+e^{-2x}} - 1$$

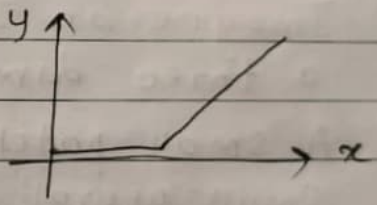
Output from -1 to 1



4) ReLU (Rectified Linear Unit)

$y = \max(0, x)$

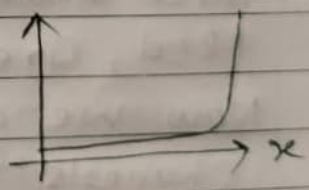
Output from 0 to ∞



5) Softmax function

Non linear

assigned with a probability



Loss Function

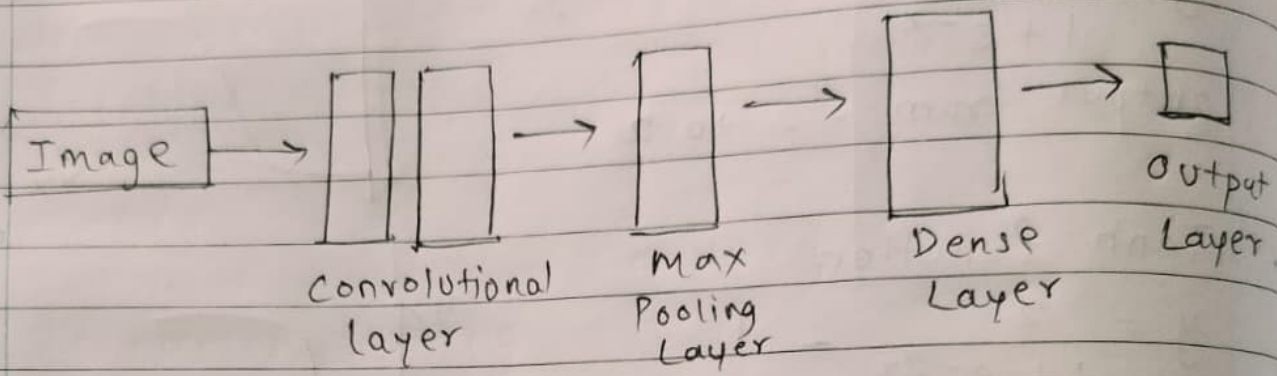
A loss function is a mathematical function that measures how well a model's prediction matches the true outcomes.

Working

1. Prediction v/s True value
2. Error measurement
3. Optimization

- Neural Network Architectures

1) Convolutional Neural Networks
→ DL network used in computer vision to understand and interpret visual data.



Working

1. Imagine an Image. It can be represented as a three dimensional structure.
2. A small patch of this image is taken and a small neural network called filter is run.
3. Three dimensions are represented as RGB (Red, Green, Blue).
4. Now we get more images with more channels (RGBs). This is convolution.
5. Visual representation of Max pooling ~~(aft~~

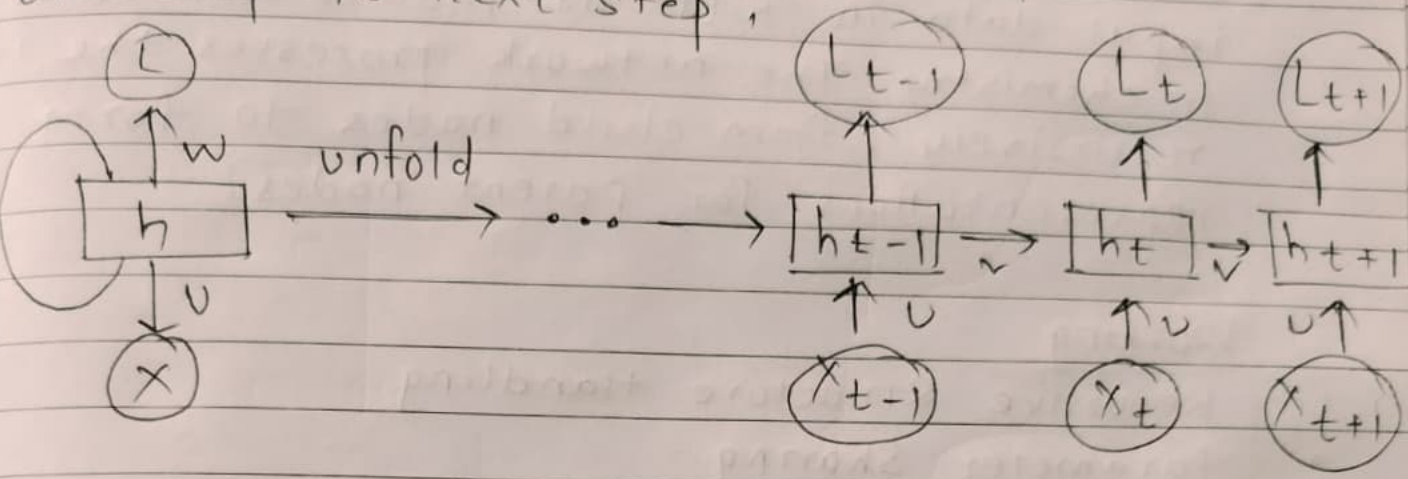
1	1	2	4
5	6	7	8
3	2	1	0
1	2	3	4

max pool 2 × 2 →

6	8
3	4

2) Recurrent Neural Networks

→ RNNs allow the network to remember past information by feeding the output from one step to next step.



working

1. Fundamental unit is a Recurrent unit that holds a hidden state. Unfolding is the process of expanding the recurrent structure over time.

2. Hidden state calculation:

$$h = \sigma(U \cdot X + W \cdot h_{t-1} + B)$$

3. Output Calculation:

$$Y = O(V \cdot h + C)$$

$O \rightarrow$ activation function

4. Overall Function:

$$Y = f(X, h, W, U, V, B, C)$$

3) Recursive Neural Networks

→ RvNN has the ability to operate on structured input data in NLP or molecular structures in chemistry. The network processes the input recursively, from child nodes to form representations for parent nodes.

Working

1. Recursive Structure Handling
2. Parameter Sharing
3. Tree Traversal
4. Composition Function

4) Unsupervised Pretrained Networks

→ UPNs train on unlabeled data to discover patterns, structures or representations within data.

components -

1) Encoder - Decoder
2) Latent space
3) Training algorithm
4) Loss Function.

5) Autoencoders.

Autoencoders learn how to compress and decompress data.

$h = f_e(W_e x + b_e)$

$f_e \rightarrow$ activation function.

An autoencoder's loss function measures how much the input data differs from its reconstruction.